

Statistical analysis of differentiable sliced optimal transport plans

Differentiable sliced optimal transport approximates the transport plan by optimizing over projection directions, but the *statistical* behaviour of the resulting estimator — how its error decays with sample size, dimension, and number of slices — is not well understood. PAC-Bayesian theory is a powerful tool here, since the average over projections is literally an average risk; but it is one tool among several (uniform convergence, concentration of the bilevel optimum). This internship analyses the generalization of differentiable sliced OT, with PAC-Bayes as the entry point.

§1. Organization

- **Start date:** As soon as possible (flexible).
- **Localization:** IRISA, Rennes.
- **Supervision:**
 - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
 - Paul Viallard (MALT Inria, IRISA, Rennes).

§2. Context

Sliced optimal transport (OT) approximates the Wasserstein distance — and, increasingly, the transport *plan* — by reducing the multivariate problem to one-dimensional projections solved in closed form. Modern variants *optimize* over the projection rather than averaging uniformly, selecting or learning a (possibly non-linear) projector that best aligns the two distributions, recovering an approximate transport plan via a differentiable bilevel formulation (Chapel et al., 2025). This sharpens the approximation but raises a statistical question: the projector is fitted to finite samples, so how does the estimator generalize, and how does its error scale with sample size n , ambient dimension d , and the number of slices L ?

Because the average over projections is literally an average risk, PAC-Bayesian theory is a natural lead tool: the distribution of slices plays the role of a posterior, and PAC-Bayes bounds control the gap between the empirical and the population estimate (Ohana et al., 2023; Alquier, 2024). It is, however, not the only tool — classical uniform-convergence and concentration arguments for sliced distances (Nadjahi et al., 2019; Nietert et al., 2022) give complementary, sometimes tighter, control — and part of the interest is to see which tool says what.

§3. Scientific challenge

A bound can serve two purposes that the literature tends to conflate. As a *training objective* it is minimized to *learn* the projector or slice distribution (the route taken by Ohana et al. (2023)); as a *certificate* it is evaluated to *guarantee* unseen-data error for a fixed estimator. Whether the sliced-OT certificate is numerically *tight* — informative rather than vacuous — is essentially uncharted, even though the analogous question reshaped PAC-Bayes for neural networks (Dziugaite & Roy, 2017). Tightness is governed by competing quantities: the KL term between the learned projector posterior and the prior, the sub-Gaussian constant of the projected transport cost (which grows with d), and the Monte-Carlo error from finitely many slices. A further subtlety is specific to the differentiable, projector-optimizing setting: the estimator is the solution of an inner optimization, so its risk inherits the randomness of that optimum, which standard sliced-OT analyses do not cover. The guiding question is: **in which (d, n, L) regimes is a generalization guarantee for differentiable sliced OT non-vacuous, and which analysis tool and design choices keep it so?**

§4. Goals

Concretely, the candidate will:

1. implement a differentiable sliced OT estimator and the associated empirical PAC-Bayes bound, reproducing a published bound as a sanity check;
2. run controlled sweeps over d , L , n , and perturbation/regularization scale, measuring the bound, the true (oracle) generalization error, and the gap between them;
3. compare data-dependent against data-independent priors, the bound used as certificate versus objective, and PAC-Bayes against at least one alternative (uniform-convergence or concentration) bound;
4. map the boundary of the non-vacuous regime and, as a stretch goal, test the protocol on a small representation-learning or transfer task.

Candidates should possess:

- a solid probability and statistics background,
- the ability to read (not necessarily re-derive) PAC-Bayes-style proofs,
- good Python coding skills for experiments,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a practical and theoretical command of generalization theory (PAC-Bayes and beyond),
- familiarity with the statistical and computational properties of sliced optimal transport,
- experience turning a theoretical bound into a rigorous experimental protocol.

§5. Bibliography

- Alquier, P. (2024). *User-friendly Introduction to PAC-Bayes Bounds*. Foundations and Trends in Machine Learning.
- Nadjahi, K., Durmus, A., Şimşekli, U., & Badeau, R. (2019). *Asymptotic Guarantees for Learning Generative Models with the Sliced-Wasserstein Distance*. NeurIPS.
- Nietert, S., Sadhu, R., Goldfeld, Z., & Kato, K. (2022). *Statistical, Robustness, and Computational Guarantees for Sliced Wasserstein Distances*. NeurIPS.
- Ohana, R., Nadjahi, K., Rakotomamonjy, A., & Ralaivola, L. (2023). *Shedding a PAC-Bayesian Light on Adaptive Sliced-Wasserstein Distances*. ICML.
- Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans*. NeurIPS. arXiv:2505.22049.
- Dziugaite, G. K., & Roy, D. M. (2017). *Computing Nonvacuous Generalization Bounds for Deep (Stochastic) Neural Networks with Many More Parameters than Training Data*. UAI.